PES-VCS Lab Report

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Course: PES VCS Lab

Phase 1: Object Storage

Command: make test_objects

Output:

```
remote: Enumerating objects: 17, done.
remote: Counting objects: 100% (17/17), done.
remote: Compressing objects: 100% (17/17), done.
remote: Total 17 (delta 0), reused 11 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (17/17), 24.80 KiB | 127.00 KiB/s, done.
```

Phase 2: Tree Objects

Command: make test_tree

Output:

```
make
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
```

Phase 3: Index (Staging Area)

Commands:

./pes init

./pes add file1.txt file2.txt
./pes status

Output:

```
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

Phase 4: Commit and Log

Commands:

./pes commit -m "Initial commit"

./pes log

Output:

```
index.c:225:5: warning: ignoring return value of ‘fread’ declared with
  225 |     fread(data, 1, size, f);
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Phase 5:

Output:

```
  229 |     if (object_write(OBJ_BLOB, data, size, &id) != 0) {
      |         ^~~~~~~~~~~~~
index.c:225:5: warning: ignoring return value of ‘fread’ declared with
  225 |     fread(data, 1, size, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Phase 6

Output:

```
cat .pes/HEAD
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Committed as a53d53d7831f2adbf44839d85b61e9e4cc159a2e190ba313fb9cf37c3fc8991b
Committed: a53d53d7831f2adbf44839d85b61e9e4cc159a2e190ba313fb9cf37c3fc8991b
a53d53d7831f2adbf44839d85b61e9e4cc159a2e190ba313fb9cf37c3fc8991b
```

Analysis Questions

Q5.1: Checkout updates HEAD to point to a branch and updates working directory files.

Q5.2: Compare index and working directory to detect modified files.

Q5.3: Detached HEAD allows commits but they are not referenced by any branch.

Q6.1: Use graph traversal (DFS/BFS) from branch heads to find reachable objects.

Q6.2: Garbage collection may delete objects in use; Git avoids this using locking and safe references.